Running ablation experiments and analyses on OpenFold for the VizFold project (December 2025):

- Document created by: Tyler Hayes
- All paths are on Delta (NCSA)
    - Project directories: bekh (ACCESS Discover) and bfna (NAIRR Pilot)
- Main code on private repo: https://github.com/tyler-hayes/vizfold
- Also located on the server here: /work/hdd/bfna/tylers_home_dir

Checking access to paths with:

ssh -t exouser@149.165.171.64 "ssh svcscigapgwuser@login.delta.ncsa.illinois.edu"

Success:

ls /projects/bekh/thayes/alignments
ls /work/hdd/bekh/openfold_outputs_layerwise_ablation
ls /work/hdd/bekh/openfold_outputs_global_ablations
ls /work/hdd/bekh/thayes/openfold_reps
ls /work/hdd/bekh/thayes/openfold_reps_outputs_sm
ls /work/hdd/bfna/cameo/layerwise_tm_csvs
ls /work/hdd/bfna/cameo/global_tm_csvs
ls /work/hdd/bfna/cameo/openfold_outputs_layerwise_ablation/
ls /work/nvme/bekh/thayes/cameo_triplets/attn_full/7ebt_B
ls /work/hdd/bfna/thayes/ecod/domain_pdbs
ls /work/hdd/bekh/thayes/ecod_dataset/fasta_files
ls /work/hdd/bfna/thayes/ecod/alignments
ls /work/hdd/bfna/thayes/ecod/experiments
ls /work/hdd/bfna/thayes/ecod/experiments/representation_saves
ls /work/hdd/bfna/thayes/ecod/experiments/openfold_reps_outputs_sm/
ls /work/hdd/bfna/thayes/FINAL_RESULT_CSVS
ls /projects/bekh/thayes/cameo_dset

Installing OpenFold Dependencies:

- Follow install instructions from the original repo
    - https://openfold.readthedocs.io/en/latest/Installation.html

AlphaFold Database Paths:

- BASE_DATA_DIR="/work/hdd/data/alphafold2/database"
- TEMPLATE_MMCIF_DIR="$BASE_DATA_DIR/pdb_mmcif/mmcif_files"
- NOTE: The old BASE_DATA_DIR was “/ime/hdd/rhaas/SUP-5301/database”. If you see this in any script, it must be replaced by "/work/hdd/data/alphafold2/database"

CAMEO PATHS:

- Truth PDB files:
  - /projects/bekh/thayes/cameo_dset/rcsb_pdb/{protein}.pdb
- FASTA FILES:
  - /projects/bekh/thayes/cameo_dset/fasta_dir/${PROT}.fasta
- MSA FILES:
  - /projects/bekh/thayes/alignments/${PROT}
- Layerwise ablation pdbs:
  - Main dir #1: /work/hdd/bekh/openfold_outputs_layerwise_ablation
    - All ablations minus the ones listed for Main dir #2
  - Main dir #2: /work/hdd/bfna/openfold_outputs_layerwise_ablation
    - Ablate Triangle Attention and TMU (FWD and REV)
    - Ablate MSA Column Attention (FWD)
- Global ablation pdbs:
  - /work/hdd/bekh/openfold_outputs_global_ablations/{PROT}/
- The 48-layer representations are here:
  - /work/hdd/bekh/thayes/openfold_reps/7ebt_B/
- The 48 structures are here (from the 48-layer representations):
  - /work/hdd/bekh/thayes/openfold_reps_outputs_sm/my_outputs_align_{PROT}_representations_load_from_saved_layer_{LAYER}/predictions/{PROT}_model_1_ptm_unrelaxed.pdb

CSV Files for CAMEO (aggregated in final CSVs, but putting them here for posterity):

- LAYERWISE TM CSVS FOR CAMEO:
  - /work/hdd/bfna/cameo/layerwise_tm_csvs
- GLOBAL TM CSVS FOR CAMEO:
  - /work/hdd/bfna/cameo/global_tm_csvs

Proteins used from CAMEO with TM-score>0.7:

['6tf4_A' '6wmk_A' '6wqc_A' '6zpp_A' '7ab3_E' '7ar0_B' '7atr_A' '7au7_A'
'7awk_A' '7b0d_A' '7b1k_B' '7b1w_F' '7b26_C' '7b28_F' '7b29_A' '7b2a_A'
'7b2o_A' '7b3a_A' '7b4q_A' '7b7t_A' '7bbz_A' '7bcz_A' '7bew_A' '7bhy_A'
'7cjs_B' '7dcm_A' '7dfe_A' '7djy_A' '7dk9_A' '7dkk_A' '7dko_A' '7dmf_A'
'7dms_A' '7dnu_A' '7don_B' '7dq9_A' '7dru_C' '7dtp_A' '7dup_A' '7dut_A'
'7dvn_A' '7e0m_A' '7e2v_A' '7e37_A' '7e3z_A' '7ebt_B' '7ee3_C' '7ef6_A'
'7eqx_A' '7esx_A' '7et8_A' '7ewu_B' '7ezg_A' '7f0h_A' '7f6e_B' '7f8a_A'
'7fbh_A' '7fbp_B' '7fe3_A' '7ffa_A' '7fh3_A' '7fip_A' '7jiz_A' '7jt9_A'
'7kdx_B' '7kik_A' '7kiu_A' '7kos_A' '7kqv_D' '7ksp_A' '7kua_A' '7kzh_A'
'7l6j_A' '7l6y_A' '7l8n_A' '7lbu_A' '7lc5_A' '7lgr_A' '7ljh_A' '7lnu_B'

'7lqm_D' '7lqn_A' '7ls0_B' '7lvz_D' '7m4n_A' '7mcc_A' '7mfi_A' '7mfw_B'
'7mpz_A' '7mrk_D' '7mro_A' '7mu9_A' '7mwr_A' '7ndr_A' '7nl4_A' '7ntn_A'
'7ny6_A' '7o49_F' '7obm_A' '7ofn_A' '7on9_A' '7ool_A' '7ouq_A' '7p3b_B'
'7p3t_B' '7p82_C' '7pab_A' '7pbk_A' '7plb_B' '7poh_B' '7ppp_A' '7pq7_A'
'7pqf_A' '7prd_A' '7puo_A' '7q03_A' '7q1b_A' '7q47_A' '7r84_A' '7rbw_A'
'7rdt_A' '7re4_A' '7re6_A' '7s94_C' '7sf6_A' '7sir_A' '7spo_C' '7spp_C'
'7stt_A' '7stz_C' '7swh_A' '7swk_B' '7sy9_A' '7t24_A' '7t5f_B' '7t71_A'
'7t9w_B' '7ta9_A' '7tav_B' '7tbs_A' '7tcb_B' '7thw_A' '7tj1_D' '7tkv_A'
'7v1v_A' '7v5y_B' '7vdy_A' '7vmu_A' '7vnb_A' '7vpf_A' '7vw0_B' '7w83_A'
'7wbr_A' '7wgk_A']

A few additional results for analyses:

- The 7ebt Pair Transition (fwd) layer ablation representations and results are here (It is 15 different experiments for different layers ablated. This is not seeded, so it might be difficult to compare runs to each other):
    - /work/hdd/bfna/cameo/openfold_outputs_layerwise_ablation/
- I saved the full triangle attention maps for 7ebt:
    - /work/nvme/bekh/thayes/cameo_triplets/attn_full/7ebt_B

ECOD PATHS:

- Selected proteins (400) with fold type and length
    - /vizfold/slurm_scripts_ecod/selected_proteins/all_categories_selected.csv
- Selected proteins as a txt file
    - /vizfold/slurm_scripts_ecod/selected_proteins/all_selected_proteins.txt
- Truth PDB files:
    - /work/hdd/bfna/thayes/ecod/domain_pdbs
- FASTA FILES:
    - /work/hdd/bekh/thayes/ecod_dataset/fasta_files/${PROT}.fasta
- MSA FILES (25000+ precomputed MSA files):
    - /work/hdd/bfna/thayes/ecod/alignments/${PROT}
- Layerwise ablation pdbs:
    - Main dir: /work/hdd/bfna/thayes/ecod/experiments
    - Example: /work/hdd/bfna/thayes/ecod/experiments/{PROT}/{prot}_ablate_msa_col_layerwise_fwd_layers_0_to_0_run1

- Global ablation pdbs:
    - Main dir: /work/hdd/bfna/thayes/ecod/experiments
    - Example: /work/hdd/bfna/thayes/ecod/experiments/{prot}/{prot}_ablate_msa_col_ablate_msa_row_global_layers_0_to_47_run1
- The 48-layer representations are here:
    - /work/hdd/bfna/thayes/ecod/experiments/representation_saves
- The 48 structures are here:
    - /work/hdd/bfna/thayes/ecod/experiments/openfold_reps_outputs_sm/my_outputs_align_{PROT}_representations_load_from_saved_layer_{LAYER}/predictions/{PROT}_model_1_ptm_unrelaxed.pdb

FULL JOINT (CAMEO AND ECOD) LAYER-WISE RESULTS IN PAPER:

- ablate_msa_col_layerwise_fwd: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_msa_col_layerwise_fwd.csv
- ablate_msa_col_layerwise_rev: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_msa_col_layerwise_rev.csv
- ablate_msa_row_layerwise_fwd: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_msa_row_layerwise_fwd.csv
- ablate_msa_row_layerwise_rev: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_msa_row_layerwise_rev.csv
- ablate_msa_transition_layerwise_fwd: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_msa_transition_layerwise_fwd.csv
- ablate_msa_transition_layerwise_rev: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_msa_transition_layerwise_rev.csv
- ablate_pair_transition_layerwise_fwd: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_pair_transition_layerwise_fwd.csv
- ablate_pair_transition_layerwise_rev: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_pair_transition_layerwise_rev.csv
- ablate_tmu_layerwise_fwd: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_tmu_layerwise_fwd.csv
- ablate_tmu_layerwise_rev: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_tmu_layerwise_rev.csv

- ablate_tri_start_ablate_tri_end_ablate_tmu_layerwise_fwd: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_tri_start_ablate_tri_end_ablate_tmu_layerwise_fwd.csv
- ablate_tri_start_ablate_tri_end_ablate_tmu_layerwise_rev: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_tri_start_ablate_tri_end_ablate_tmu_layerwise_rev.csv
- ablate_tri_start_ablate_tri_end_layerwise_fwd: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_tri_start_ablate_tri_end_layerwise_fwd.csv
- ablate_tri_start_ablate_tri_end_layerwise_rev: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/ablate_tri_start_ablate_tri_end_layerwise_rev.csv
- structure_module_tm_scores: 554 total rows (ECOD=400, CAMEO=154)
    - /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/structure_module_tm_scores.csv

FULL JOINT (CAMEO AND ECOD) GLOBAL RESULTS IN PAPER:

- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_msa_col_ablate_msa_row_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_msa_col_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_msa_row_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_msa_transition_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_pair_transition_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_tmu_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_tri_start_ablate_tri_end_ablate_msa_col_ablate_msa_row_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)

- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_tri_start_ablate_tri_end_ablate_msa_col_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_tri_start_ablate_tri_end_ablate_msa_row_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_tri_start_ablate_tri_end_ablate_tmu_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__ablate_tri_start_ablate_tri_end_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__baseline_global_layers_0_to_47.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__zero_m_before_sm.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__zero_z_before_sm.csv (ECOD=400, CAMEO=154, total=554)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/global__baseline_numrec3.csv (ECOD=400, CAMEO=0, total=400)
- /work/hdd/bfna/thayes/FINAL_RESULT_CSVS/global/MASTER_global_all_experiments.csv

Important Scripts (using ECOD as reference, but same scripts apply to CAMEO if paths are changed):

- Main inference script for OpenFold used in all experiments:
  - /vizfold/run_pretrained_openfold.py
- Compute MSAs
  - /vizfold/slurm_scripts_ecod/run_alignment_job.sh
    - Submission script: /vizfold/slurm_scripts_ecod/run_alignment_jobs_missing.sh (provide text file of IDs to run)
- Compute 3-recycle baseline
  - /vizfold/slurm_scripts_ecod/baseline_3_recycles.sh
    - Submission script: /vizfold/slurm_scripts_ecod/launch_baseline_3_recycles.sh
- Compute 0-recycle baseline
  - /vizfold/slurm_scripts_ecod/baseline_0_recycles.sh
    - Submission script: /vizfold/slurm_scripts_ecod/launch_baseline_3_recycles.sh
- Save 48-layer representations to disk
  - /vizfold/slurm_scripts_ecod/baseline_representation_saves.sh
    - Submission script: /vizfold/slurm_scripts_ecod/submit_baseline_representation_saves.sh

- Compute structures per layer from saved representations
    - Note: need to run code to save 48-layer representations to disk first
    - /vizfold/slurm_scripts_ecod/inference_representation_load_from_layer.sh
        - Submission script: /vizfold/slurm_scripts_ecod/submit_inference_representation_load_from_layer.sh
- Run global ablations of components in the Evoformer:
    - /vizfold/slurm_scripts_ecod/global_ablations_components.sh
        - Submission script: /vizfold/slurm_scripts_ecod/submit_global_ablations_components_sub.sh (provide text file of IDs to run)
- Run global ablation where we zero MSA or Pair representation before the Structure Module
    - /vizfold/slurm_scripts_ecod/global_ablations_representations.sh
        - Submission script: /vizfold/slurm_scripts_ecod/submit_global_ablations_representations_sub.sh (provide text file of IDs to run)
- Run layer ablations of components in the Evoformer
- /vizfold/slurm_scripts_ecod/inference_general_per_layer_ablate_flags_LAYER_UPDATE.sh
    - Submission script: /vizfold/slurm_scripts_ecod/submit_per_layer_ablate_flags_LAYER_UPDATE.sh
- Example scripts for computing TM-scores from experiments:
    - Global ablations: /vizfold/slurm_scripts_ecod/compute_tm_scores.py and .sh
    - Layer ablations: /vizfold/slurm_scripts_ecod/compute_tm_scores_LAYERWISE.py and .sh
    - Note: to compute LDDT, use Openstructure from conda
- If you want to ablate more than one component in various layers, use this script as an example (from CAMEO)
    - This script can also show how to make the dictionary of layers for various layer ablations
    - /vizfold/slurm_scripts_cameo/inference_general_MINIMUM_SET_DEC_2025.sh
        - Submission script: /vizfold/slurm_scripts_cameo/submit_inference_general_MINIMUM_SET_DEC_2025.sh

Scripts to make final combined CSV files and plot final plots for paper:

- Script to compile file CSV files with all results (mostly just here as a reference for paths and naming of results)
    - /vizfold/slurm_scripts_cameo/MAKE_FINAL_CSVS_GLOBAL_JOINT_DEC.py

    - /vizfold/slurm_scripts_cameo/MAKE_FINAL_CSVS_LAYER_JOINT_DEC.py
- Scripts to make final ablation plots
    - Violin plots of Global Ablations: /vizfold/slurm_scripts_cameo/MAKE_FINAL_GLOBAL_PLOTS_JOINT_DEC.py
    - Plots of Layer-wise Ablations: /vizfold/slurm_scripts_cameo/MAKE_FINAL_LAYER_PLOTS_JOINT.py
- Scripts to compute Cosine Similarity and CKA of representations from layer L to final layer
    - Compute statistics: /vizfold/slurm_scripts_cameo/MAKE_FINAL_CORR_COS_REP_DEC.py and .sh
    - Make plots: /vizfold/slurm_scripts_cameo/MAKE_FINAL_CORR_COS_PLOTS.py
- Scripts to cluster layer-wise ablation results
    - Compute clustering: /vizfold/slurm_scripts_cameo/MAKE_FINAL_LAYER_CLUSTER.py
    - Make "Switch Rate" plot: /vizfold/slurm_scripts_cameo/MAKE_FINAL_LAYER_CLUSTER_PLOTS_ARI.py
- Script to make plots of subtracting FWD layer-wise results and REV layer-wise results
    - /vizfold/slurm_scripts_cameo/MAKE_FINAL_LAYER_SUBTRACT.py

For saving attention maps, I'd suggest looking at the cleaner demo repo:

- https://github.com/vizfold/attention-viz-demo